

EX
1

Développer et réduire les expressions suivantes.

1. $(x - 5)(x + 5)$

2. $(x - 8)^2$

3. $(x + 7)^2$

4. $(x - 9)(x + 9)$

5. $(x - 7)^2$

6. $(x + 8)^2$

7. $(x + 6)^2$

8. $(x - 8)(x + 8)$

9. $(x - 5)^2$

EX
2

Développer et réduire les expressions suivantes.

1. $(5x - 3)^2$

2. $(6x + 3)^2$

3. $(5x - 9)(5x + 9)$

4. $(5x - 8)(5x + 8)$

5. $(6x + 9)^2$

6. $(2x - 2)^2$

7. $(6x + 7)^2$

8. $(2x - 6)^2$

9. $(5x - 2)(5x + 2)$

EX
3

Factoriser les expressions suivantes.

$A = 4x^2 - 9$

$B = 4x^2 - 64$

$C = 64x^2 - 25$

$D = 9x^2 - 49$

$E = 81x^2 - 1$

$F = 4x^2 - 36$

$G = 16x^2 - 4$

$H = 36x^2 - 81$

$I = 4x^2 - 16$



EX

1

1. $(x-5)(x+5) = x^2 - 5^2 = x^2 - 25$
2. $(x-8)^2 = x^2 - 2 \times 8 \times x + 8^2 = x^2 - 16x + 64$
3. $(x+7)^2 = x^2 + 14x + 49$
4. $(x-9)(x+9) = x^2 - 9^2 = x^2 - 81$
5. $(x-7)^2 = x^2 - 2 \times 7 \times x + 7^2 = x^2 - 14x + 49$
6. $(x+8)^2 = x^2 + 16x + 64$
7. $(x+6)^2 = x^2 + 12x + 36$
8. $(x-8)(x+8) = x^2 - 8^2 = x^2 - 64$
9. $(x-5)^2 = x^2 - 2 \times 5 \times x + 5^2 = x^2 - 10x + 25$

EX

2

1. $(5x-3)^2 = (5x)^2 - 2 \times 5x \times 3 + 3^2 = 25x^2 - 30x + 9$
2. $(6x+3)^2 = (6x)^2 + 2 \times 6x \times 3 + 3^2 = 36x^2 + 36x + 9$
3. $(5x-9)(5x+9) = (5x)^2 - 9^2 = 25x^2 - 81$
4. $(5x-8)(5x+8) = (5x)^2 - 8^2 = 25x^2 - 64$
5. $(6x+9)^2 = (6x)^2 + 2 \times 6x \times 9 + 9^2 = 36x^2 + 108x + 81$
6. $(2x-2)^2 = (2x)^2 - 2 \times 2x \times 2 + 2^2 = 4x^2 - 8x + 4$
7. $(6x+7)^2 = (6x)^2 + 2 \times 6x \times 7 + 7^2 = 36x^2 + 84x + 49$
8. $(2x-6)^2 = (2x)^2 - 2 \times 2x \times 6 + 6^2 = 4x^2 - 24x + 36$
9. $(5x-2)(5x+2) = (5x)^2 - 2^2 = 25x^2 - 4$

EX

3

1. $A = 4x^2 - 9$
 $A = (2x)^2 - 3^2$
 $A = (2x-3)(2x+3)$
2. $B = 4x^2 - 64$
 $B = (2x)^2 - 8^2$
 $B = (2x-8)(2x+8)$
3. $C = 64x^2 - 25$
 $C = (8x)^2 - 5^2$
 $C = (8x-5)(8x+5)$
4. $D = 9x^2 - 49$
 $D = (3x)^2 - 7^2$
 $D = (3x-7)(3x+7)$
5. $E = 81x^2 - 1$
 $E = (9x)^2 - 1^2$



$$E = (9x - 1)(9x + 1)$$

6. $F = 4x^2 - 36$

$$F = (2x)^2 - 6^2$$

$$F = (2x - 6)(2x + 6)$$

7. $G = 16x^2 - 4$

$$G = (4x)^2 - 2^2$$

$$G = (4x - 2)(4x + 2)$$

8. $H = 36x^2 - 81$

$$H = (6x)^2 - 9^2$$

$$H = (6x - 9)(6x + 9)$$

9. $I = 4x^2 - 16$

$$I = (2x)^2 - 4^2$$

$$I = (2x - 4)(2x + 4)$$